

RAP 305 Test Equipment

From Rovipedia, the Rovisys encyclopedia

Approved

Purpose: This procedure covers the purchase, calibration, tracking, and use of test equipment. The RoviSys Company uses many test instruments to test designs, verify system operation and calibrate customer equipment. Care and calibration of our test equipment is important to achieve our goal of well designed and accurate systems and products. These guidelines are to be followed, when appropriate, to achieve this end result.

Definitions

Test Equipment: Equipment used by RoviSys to verify levels of voltage, current, pressure, temperature or other engineering units.

Calibration Equipment: Equipment used by RoviSys to calibrate customer or internal measurement devices (thermocouples, RTDs, or other sensing devices) in engineering units such as voltage, current, and pressure.

Calibration Cycle: The frequency that equipment is subject to external verification of accuracy and/or traceability to standards.

Lab Equipment Coordinator: Engineer or Senior Technician responsible for the checkout and lab area, tools, and test equipment. Each RoviSys office with a checkout facility will designate an individual responsible for all lab equipment. The main office will take control of the master list, and the Lab Equipment Coordinator for that site will work through their counterparts at any remote sites to ensure that this RAP is consistently applied.

1. Purchase of equipment for use by The RoviSys Company shall be reviewed by knowledgeable engineers and the Engineering Manager for suitability to task. Overall accuracy, repeatability, linearity, and traceability to standards shall be reviewed relative to the specific requirements for the instrument being purchased. Other issues such as ease of use, size and weight, etc., may also be considered on some equipment purchases.
2. Each piece of RoviSys test equipment shall be logged with information including: vendor name, part number, serial number, Capital Asset Number, if cost is greater than \$500, and any other information necessary to uniquely designate each piece of equipment. This database shall be maintained by the Lab Equipment Coordinator and is accessible from GrandView.
3. When the RoviSys Company purchases equipment, the Lab Equipment Coordinator shall post a status log in GrandView notifying all employees that the equipment is available for use.
4. Equipment calibration cycle will be determined by the Lab Equipment Coordinator, based on vendor recommendations and RoviSys use. Typical cycle for most solid state electronics is 12 months. Cycle duration will be entered into the test equipment database. Monthly reports will be generated to indicate any piece of test equipment that is due for calibration.
5. The Lab Equipment Coordinator will review monthly calibration reports and is responsible for ensuring that all equipment is kept within the scheduled calibration cycle or is taken from circulation.
6. Any test equipment that is on a calibration cycle must be signed out when removed from the cabinet. The name of the person using the equipment, project used for, projected dates of use, and the office or site where the equipment is to be kept will be recorded when the equipment is signed out.
7. If a piece of equipment that is signed out and in use becomes due for calibration, the Lab Equipment Coordinator shall notify the Project Engineer to whom it is signed out. It is up to the project engineer to

- ensure that sufficient test equipment is available for his/her project and that the calibration is up to date on all equipment. Project Engineers may request early calibration to avoid interruption of service when, for instance, calibration is due during the scheduled period of use.
8. Equipment may be calibrated by internal standards, outside test labs or other appropriate means. The calibration shall use test gear and standards that are more accurate than the equipment being calibrated. Traceability to NIST or other standards may or may not be necessary, as determined by RoviSys management and Project Engineers according to the intended use of the equipment. Customers may request specific traceability and shall receive it when agreed to as a condition to sale or negotiated with the RoviSys Project Manager.
 9. If the calibration is carried out by an external party, the Lab Equipment Coordinator shall obtain a quote to be approved by Quality Assurance. The Lab Equipment Coordinator shall then complete a Purchase Order for the calibration.
 10. Calibration shall include a certificate of calibration stating the accuracy attained, traceability to standards, date of calibration, and person/vendor doing the calibration. A Calibration Due Sticker shall be attached to each piece of equipment that indicates when it was calibrated and when it is due for calibration.
 11. Lab book references to calibration equipment should include the equipment designation, so that it may be traced against the calibration database maintained by the Lab Equipment Coordinator.

Page History

Revision	Action/Comment	Date	User
r12 (this)	Approve	7-Mar-2023 2:05 UTC	Austin Beamish
r12 (this)	Create/Edit <i>Reviewed by Eric Lake, Lab Equip Coor...</i>	6-Mar-2023 8:52 UTC	Beverly Kiglics
r11	Approve	24-Aug-2020 4:58 UTC	Jeff Marshall
r11	Create/Edit <i>Fixed RoviSys to RoviSys</i>	6-Aug-2020 3:56 UTC	Shereef Elbarbary
r10	Approve	6-Jun-2018 5:22 UTC	John Turk
r10	Create/Edit <i>Changed ì to i in RoviSys</i>	4-Jun-2018 8:59 UTC	Travis Bolick
r9	Create/Edit <i>Changed ì to i in RoviSys</i>	4-Jun-2018 8:59 UTC	Travis Bolick
r8	Approve	11-Jan-2013 12:36 UTC	Jeff Marshall
r8	Create/Edit <i>Fixed tags</i>	10-Jan-2013 12:21 UTC	Dave Hermann
r7	Approve	17-Aug-2012 3:24 UTC	Pete Madonia
r7	Create/Edit <i>Slight grammatical and capitalization...</i>	30-May-2012 3:39 UTC	Ruoth, Joel
r6	Approve	7-Mar-2011 2:16 UTC	John Turk
r6	Create/Edit <i>Imported rev. cf of RAP into RoviPedia</i>	2-Sep-2010 6:45 UTC	Tom Grulkowski
r5	Create/Edit <i>updated tags</i>	27-Aug-2010 2:10 UTC	Haren, Michael
r4	Create/Edit <i>updated tags</i>	27-Aug-2010 3:51 UTC	Haren, Michael
r3	Create/Edit <i>Formatting</i>	25-Aug-2010 6:55 UTC	Xander Dumaine
r2	Create/Edit <i>Formatting</i>	25-Aug-2010 6:54 UTC	Xander Dumaine
r1	Create/Edit <i>Initial revision</i>	25-Aug-2010 6:47 UTC	Xander Dumaine

Note: This is a snapshot of a living document. Please check online for the latest revision.

Edited: 6-Mar-2023 8:52 UTC **Printed:** 20-Oct-2023 6:52 UTC

Document Number: 99 **Revision:** r12